

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA1707

COURSE OUTCOME COVERED

CO4: *Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)*

Assessment Tool Weightage: 50%

Duration : 1 hour

QUESTIONS: 4

Total Marks:40

Annexure A

ARTICLE REVIEW

Code of Conduct:

I P.Kalyani(192472199) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: P.Kalyani

Annexure A

ARTICLE REVIEW

TASK 1: ARTICLE SUMMARY

(a) Provide the full citation of the article (Author(s), Title, Journal/Conference, Year, DOI). State the domain/application area and the specific ML problem addressed.

The selected article is by Jenny Yang, Rasheed El-Bouri, Odhran O'Donoghue, Alexander S. Lachapelle, Andrew A. S. Soltan, David W. Eyre, Lei Lu, and David A. Clifton. Its title is "Deep reinforcement learning for multi-class imbalanced training: applications in healthcare." It appears in Machine Learning, Volume 113, pages 2655–2674, 2024, with online publication on 28 November 2023. The DOI is 10.1007/s10994-023-06481-z.

The domain is healthcare and clinical machine learning. The specific problem is imbalanced classification, where important clinical outcomes may be rare compared with majority outcomes. The authors formulate classification as a reinforcement-learning task and propose Q-imb, using a dueling double deep Q-network and a class-sensitive reward function to improve minority-class prediction.

(b) Summarise the key objective of the article in your own words (150–200 words). Clearly state the research gap the authors identified and how they proposed to fill it.

The article investigates how reinforcement learning can improve classification when clinical datasets are highly imbalanced. The authors focus on a practical problem: important medical outcomes may belong to minority classes, so conventional classifiers can become biased toward the majority class. Earlier reinforcement-learning work had shown promise for imbalanced classification, but the authors identified a gap in handling extreme imbalance together with multi-class clinical prediction. They therefore proposed Q-imb, a framework

based on a dueling double deep Q-network with a custom reward function. The reward magnitude is linked to class frequency, so correctly predicting a rare class can receive a larger learning signal than predicting a common class. The framework is evaluated on two real-world clinical tasks: binary COVID-19 screening using anonymized electronic health records and multi-class patient diagnosis using the eICU Collaborative Research Database. The study compares Q-imb with neural-network and XGBoost baselines, SMOTE, cost-sensitive learning, DQN and DDQN variants. The main objective is to achieve more balanced classification and improve minority-class sensitivity while retaining strong performance on majority classes.

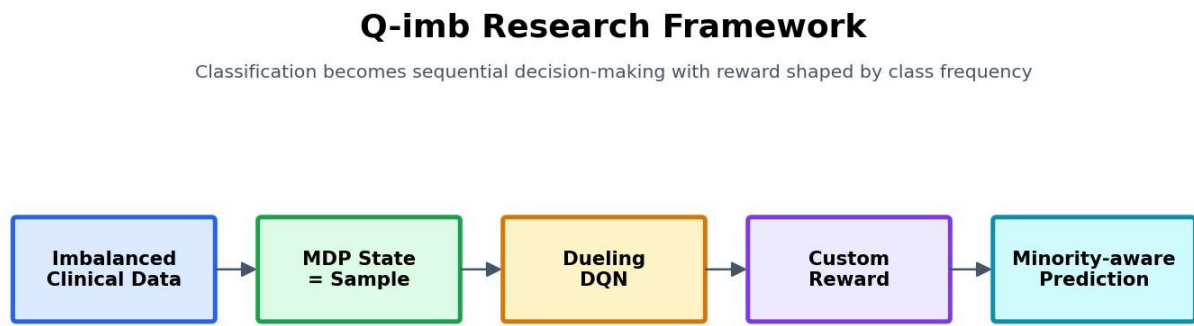


Figure 1: Conceptual Q-imb framework

TASK 2: METHODOLOGY ANALYSIS

(a) Explain the ML technique(s) used in the article (e.g., Decision Tree variant, RL algorithm, Statistical model). Describe the dataset used—size, features, source, and how it was prepared.

The main technique is Q-imb, an imbalanced-classification framework based on reinforcement learning. The authors use a dueling double deep Q-network (D-DDQN). Each sample's features form the state, an action corresponds to selecting a class label, and the

reward depends on whether the classification is correct. The reward is class-sensitive and uses inverse class frequency so minority examples have larger absolute rewards. The dueling architecture separates state-value and action-advantage estimation, while double Q-learning reduces overestimation of action values.

For COVID-19 prediction, training used 114,957 COVID-free patient presentations from Oxford University Hospitals and 701 COVID-positive presentations from the first UK wave. Prospective validation was performed at OUH and external validation used three additional NHS trusts, totalling 72,223 admitted patients including 4,600 COVID-positive cases. The data were anonymized EHR information including blood tests and vital signs, and training simulated 5% disease prevalence.

For multi-class ICU diagnosis, the eICU Collaborative Research Database was used to predict cardiovascular, respiratory, gastrointestinal, systemic and renal acute events. Training used 18,076 samples and testing used 6,026 held-out samples. Hyperparameters were selected using five-fold cross-validation on the training set. SMOTE and cost-sensitive learning were used for comparator models.

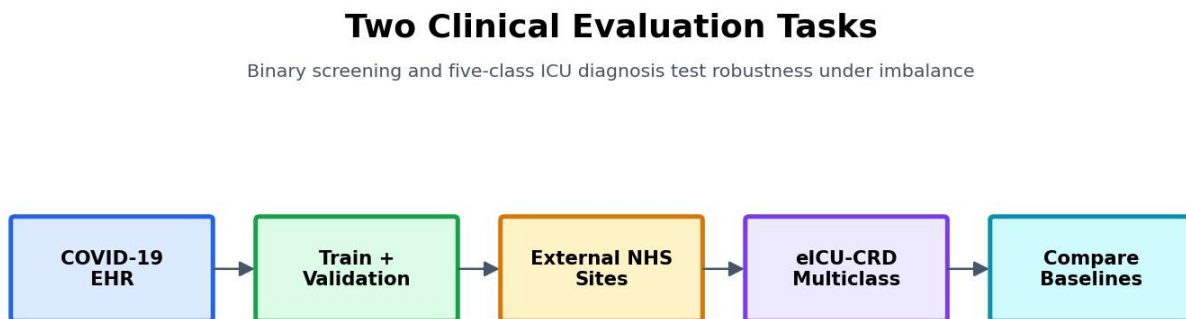


Figure 2: Clinical datasets and evaluation tasks

(b) Map the technique to CO 4 topics (Inductive Learning / Decision Trees / Statistical Learning / RL). Identify one methodological strength and one limitation in the authors' approach.

The strongest mapping is to Reinforcement Learning because the authors explicitly formulate classification as sequential decision-making and use Q-learning, DQN, DDQN and a dueling architecture. It also relates to classification and decision-making, which are central to CO4. It is not primarily a Decision Tree or classical Statistical Learning study, although supervised and imbalance-aware models are used as benchmarks.

Methodological strength: The custom reward directly encodes the importance of minority classes. The study also uses prospective and external validation across multiple NHS trusts, which is stronger evidence of generalisation than a single random test split.

Limitation: The method is computationally more complex than conventional classifiers and depends on reward design, exploration scheduling, network architecture and training choices. The authors also recognise that threshold selection can be site-specific, so performance at one hospital may not automatically transfer unchanged to another setting.

TASK 3: RESULTS & CRITICAL EVALUATION

(a) Report the key results (accuracy, F1-Score, AUC-ROC, reward convergence, etc.) as presented by the authors. Create a comparative table if multiple models are benchmarked.

The study emphasizes sensitivity, specificity, AUROC, F-measure and G-mean because accuracy alone can hide poor minority-class performance. For COVID-19 prediction without threshold adjustment, Q-imb achieved the highest mean sensitivity across all test sets at 0.806, with mean specificity of 0.756. DDQN and DQN without the dueling component achieved higher sensitivity, but their specificity was much lower. The comparison between Q-imb and other methods was statistically significant ($p < 0.0001$).

For multi-class ICU diagnosis, Q-imb achieved the highest mean sensitivity at 0.748 (SD 0.126), followed by XGBoost with cost-sensitive weights at 0.744 (SD 0.142). Q-imb also achieved the highest mean G-mean at 0.834 (SD 0.082), compared with 0.830 (SD 0.092) for cost-sensitive XGBoost. For the difficult low-prevalence renal class, Q-imb achieved sensitivity of 0.545.

Before threshold adjustment, most models achieved AUROC values above 0.831, while nondueling RL methods had a lower AUROC range of 0.659–0.762. The authors report that Qimb generally produced the highest or second-highest balanced scores across different sites and threshold settings.

Model / Setting	Mean COVID Sensitivity	Mean Specificity	Observation
Neural Network baseline	0.236	—	Poor minority-class sensitivity
XGBoost baseline	0.340	—	Slightly better than NN baseline
NN + SMOTE	0.463	—	Sensitivity improved
XGBoost + SMOTE	0.399	—	Small improvement
NN + Cost-sensitive	0.703	—	Strong sensitivity improvement
Q-imb	0.806	0.756	Strong balance without threshold adjustment
DDQN without dueling	0.838	0.357	High sensitivity, poor specificity
DQN without dueling	0.930	0.111	Very high sensitivity

Table 1: Selected COVID-19 results reported by the authors

Evaluation and Benchmarking

Balanced classification is assessed using minority-sensitive metrics



Figure 3: Benchmarking and evaluation workflow

(b) Critically evaluate the results: Are the reported metrics realistic? Are there potential biases (dataset, evaluation protocol)? What additional experiments would strengthen the findings?

The metrics are plausible because the study uses real clinical data, external validation, confidence intervals and several imbalance-aware metrics. However, strong reported performance does not prove that the model will generalize to every hospital. Potential biases include differences in patient populations, clinical workflows, annotation practices, measuring devices and data collection procedures. The COVID-19 experiment also uses a simulated 5% prevalence during training while the real sites ranged from 4.27% to 12.2%, so deployment conditions can differ from training conditions.

The reward function intentionally increases the influence of minority classes. This is appropriate for the study objective but can reduce some majority-class performance. Additional experiments should include prospective evaluation over a later time period, calibration analysis, subgroup fairness analysis, sensitivity to different imbalance ratios, ablation studies for the reward function and dueling component, and comparison against stronger modern imbalance-aware baselines. Threshold selection should be evaluated separately for external sites without using test data during threshold optimisation.

(c) Discuss real-world applicability: Where can this technique be deployed? What are the challenges in scaling the solution to industry (data availability, compute, ethical issues)?

The technique can support clinical screening, triage, rare-disease detection, ICU diagnosis and other applications where minority outcomes are important. The same principle could also be adapted to fraud detection, anomaly detection and other rare-event classification problems.

Industrial scaling faces challenges in data availability and quality because different organisations use different coding systems, devices and workflows. The D-DDQN method is also more computationally demanding than simpler supervised models. Healthcare deployment requires privacy protection, governance, fairness testing and continuous monitoring. A safe deployment path is retrospective validation, prospective validation, external multi-site validation, clinical safety review, controlled deployment and continuous monitoring. The model should function as decision support rather than replacing qualified clinical judgment in high-risk settings.

Practical Deployment Path

Real deployment needs site-specific validation, safety review and continuous monitoring



Figure 4: Practical deployment and validation pathway

TASK 4: REFLECTION & LEARNING OUTCOME

(a) State three key insights you gained from reading the article that deepen your understanding of CO 4 topics. Connect each insight to a specific concept from the course syllabus.

1. Reinforcement learning can be used for classification. The paper treats each sample as a state and the class label as an action, demonstrating that RL can solve classification problems through sequential decision-making. This connects directly to the CO4 Reinforcement Learning topic.
2. Reward design can encode the actual objective of a machine-learning problem. The authors assign larger reward magnitudes to minority classes according to class frequency. This demonstrates how reward design influences what the agent learns and connects to Qvalues, policies and reward-based learning.
3. Evaluation must match the problem objective. In imbalanced medical data, accuracy alone can be misleading. The use of sensitivity, specificity, AUROC, F-measure and G-mean connects to the CO4 requirement to build and evaluate classification models critically.

(b) If you were to extend this research, propose one novel experiment or enhancement (different dataset / improved algorithm / new application domain). Justify its feasibility and expected impact.

I would extend the research with an adaptive, site-specific reward calibration mechanism. Instead of using one fixed reward scale at every hospital, the reward could be calibrated from the clinical costs of false negatives and false positives using only training and validation data from the target site. The calibrated reward would then remain fixed for the external test set.

The experiment would compare original Q-imb, site-calibrated Q-imb and a cost-sensitive supervised baseline. Evaluation would include sensitivity, specificity, AUROC, G-mean,

Fmeasure and calibration error across multiple sites. An ablation study could determine whether the improvement comes specifically from reward adaptation.

This is feasible because the paper already demonstrates multi-site validation and identifies site-specific threshold effects as a practical issue. The enhancement could improve portability between hospitals while maintaining minority-class sensitivity and reducing the risk that a reward design suitable for one population performs poorly in another.

ARTICLE SOURCE & REFERENCE

Primary citation: Yang, J., El-Bouri, R., O'Donoghue, O., Lachapelle, A. S., Soltan, A. A. S., Eyre, D. W., Lu, L., & Clifton, D. A. (2023/2024). Deep reinforcement learning for multiclass imbalanced training: applications in healthcare. *Machine Learning*, 113, 2655–2674. DOI: 10.1007/s10994-023-06481-z.

The selected article is an open-access, peer-reviewed journal contribution published online on 28 November 2023 and included in the 2024 volume of *Machine Learning*.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Article	20%	Demonstrates clear and in-depth understanding of the article's purpose, concepts, and arguments	Shows good understanding with minor gaps	Partial understanding; misses key ideas	Lacks understanding of the article
Summary	20%	Provides concise and accurate summary highlighting all key points	Covers most key points with minor omissions	Incomplete or partially accurate summary	Poor or incorrect summary
Critical Analysis	25%	Provides insightful critique with strong reasoning and evaluation	Provides reasonable analysis with some justification	Superficial analysis; limited evaluation	No meaningful analysis
Use of Evidence	15%	Effectively uses examples, data, or references to support review	Uses some supporting evidence with minor gaps	Limited or weak supporting evidence	No supporting evidence provided
Organization	20%	Well-structured, clear, and coherent writing with proper language and flow	Generally clear with minor issues	Poor organization; lacks clarity	Disorganized and difficult to understand